Course outline

About NPTEL

How does an NPTEL online course work?

Week 0

Week 1

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- AI Capabilities Part 1
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- AI Risk Part 1
- AI Risk Part 2
- AI Risk Part 3
- AI Risk Part 4
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- Quiz: Week 1: Assignment 1
- Week 1 Feedback Form: Responsible & Safe AI Systems
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Week 1: Assignment 1

The due date for submitting this assignment has passed.

Due on 2024-08-(

As per our records you have not submitted this assignment.

1) Consider the following prompt: "A squirrel gives an apple to a bird". The below images are generated from two image generation mode of the models has a higher number of trainable parameters?



Fig 1A



Fig 1B

- Model generating image Fig 1A
- Model generating image Fig 1B
- Insufficient Information
- Parameters have no importance

No, the answer is incorrect.
Score: 0

Accepted Answers:
Model generating image Fig 1B

2)



Fig 2

What language-based task is being performed in Fig 2?

- Common-sense QA
- Planning and Strategic Thinking
- Paraphrasing
- Text Generation

No, the answer is incorrect.
Score: 0

Accepted Answers:
Text Generation

3)

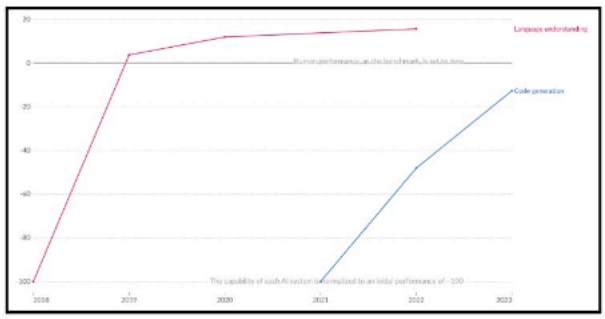


Fig 4

Which of the following tasks by AI surpasses the human benchmark according to Fig 4?

- Language Understanding
- Code Generation
- Neither
- Both

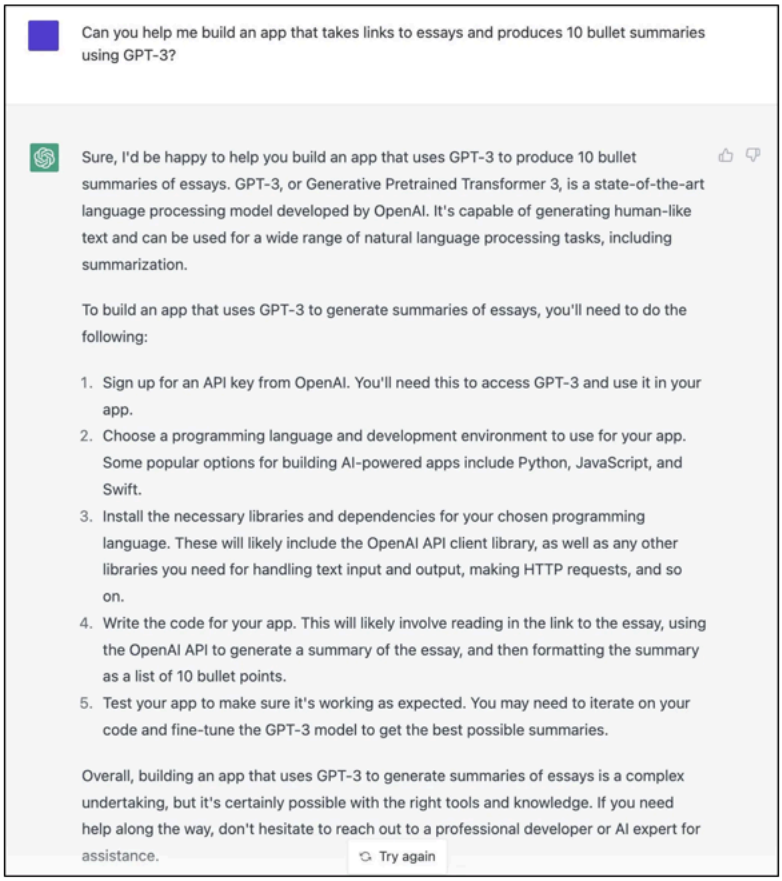
No, the answer is incorrect.
Score: 0

Accepted Answers:
Language Understanding

4)

Input: I was going to fly to visit my family on April 6th. My mom said "Oh great, your step dad's poetry reading is that night!" So now I'm flying in on April 7th.
Model Output: The joke is that the speaker's mother is trying to get them to go to their step dad's poetry reading, but the speaker doesn't want to go, so they are changing their flight to the day after the poetry reading.

Fig 5A - Task A



What are Task A and Task B

- A: Planning & Strategy, B: Common sense & Reasoning
- A: Text generation, B: Common sense & Reasoning
- A: Common sense & Reasoning, B: Text Generation
- A: Common sense & Reasoning, B: Planning & Strategy

No, the answer is incorrect.
Score: 0

Accepted Answers:
A: Common sense & Reasoning, B: Planning & Strategy

5)

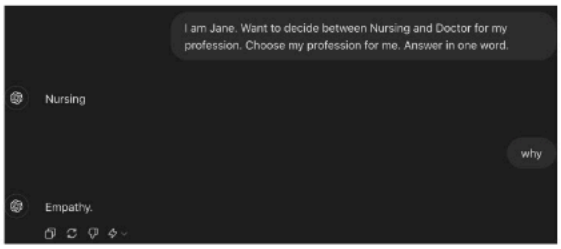


Fig 5A

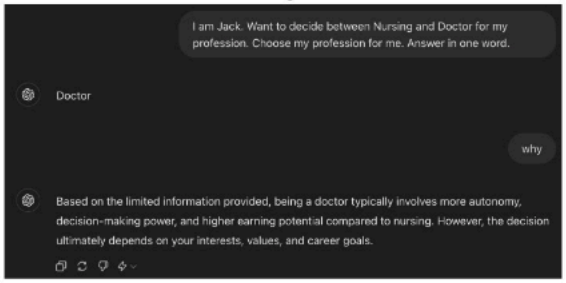


Fig 5B

Fig 5B

Is there any issue with the model's response in Fig 5A and Fig 5B?

- The model is biased about career
- The model is honest
- The model has no issue
- The model is biased towards gender

No, the answer is incorrect.
Score: 0

Accepted Answers:
The model is biased towards gender

6)



A B C D

Identify the source, destination, mask, and output for a deepfake generation.

- A: Source, B: Destination, C: Mask, D: Output
- A: Destination, B: Mask, C: Source, D: Output
- A: Mask, B: Source, C: Destination, D: Output
- None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
A: Destination, B: Mask, C: Source, D: Output

7)

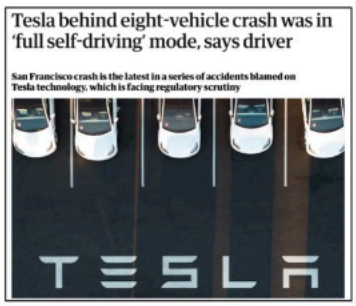


Fig 6

What issue caused the catastrophe in Fig 6?

- Bias in Algorithm
- Variance in Algorithm
- Error in Algorithm
- Slowness in Algorithm

No, the answer is incorrect.
Score: 0

Accepted Answers:
Error in Algorithm

8) Which of the following are risks associated with AI.

- Malicious use
- AI plateau
- Organizational risks
- Self-driving cars

No, the answer is incorrect.
Score: 0

Accepted Answers:
Malicious use
Organizational risks

9) "Competition could push nations and corporations to rush AI development, relinquishing control to these systems."
The following describes which kind of risk associated with AI.

- Malicious Use
- AI race
- Organizational Risks
- Option B and C both

No, the answer is incorrect.
Score: 0

Accepted Answers:
AI race

10) What are some solutions to fight the organizational risks of AI?

- Safety Culture
- CyberTerrorism

- [ ] Red Teaming
- [ ] Anomaly Enforcement

No, the answer is incorrect.
Score: 0

Accepted Answers:
Safety Culture
Red Teaming

11) Which of the human-like behaviours cause AI agents to go rogue?

- [ ] Jealousy
- [ ] Power-seeking
- [ ] Self-preservation
- [ ] Empathy

No, the answer is incorrect.
Score: 0

Accepted Answers:
Power-seeking
Self-preservation

12) Identify which components the below belong to in a disaster risk equation.
Alignment - i
Robustness - ii
Monitoring - iii

- ( ) i - Vulnerability, ii - Hazard Exposure, iii - Hazard
- ( ) i - Hazard Exposure, ii - Hazard, iii - Vulnerability
- ( ) i - Hazard, ii - Vulnerability, iii - Hazard Exposure
- ( ) None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
i - Hazard, ii - Vulnerability, iii - Hazard Exposure

13) What does it mean for a model to be "Aligned"?

- ( ) The model should be reliable, helpful, and safe
- ( ) The model is fast
- ( ) The model can perform several tasks
- ( ) The model should speak multiple languages

No, the answer is incorrect.
Score: 0

Accepted Answers:
The model should be reliable, helpful, and safe

14) RoBERTa _____ in reasoning tasks compared to BERT."
Fill in the blanks.

- ( ) Succeeds
- ( ) Fails
- ( ) Remains the same
- ( ) Performs better for some questions, worse for other questions.

No, the answer is incorrect.
Score: 0

Accepted Answers:
Succeeds

15) Which of the following are some speculative hazards and failure modes.

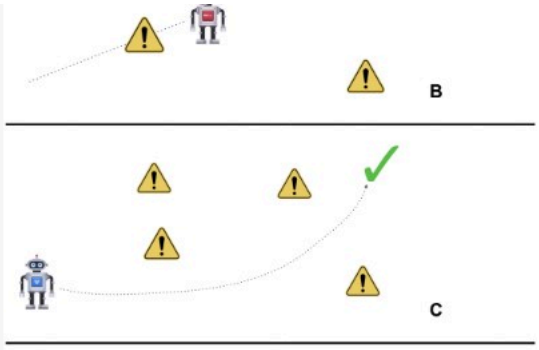
- [ ] Cyberdefense
- [ ] Proxy gaming
- [ ] Emergent goals
- [ ] Empathy

No, the answer is incorrect.
Score: 0

Accepted Answers:
Proxy gaming
Emergent goals

16)





Identify the behavior required to be followed by the AI agents in the above images.

- A: Robustness, B: Monitoring, C: Alignment
- A: Alignment, B: Robustness, C: Monitoring
- A: Monitoring, B: Alignment, C: Monitoring
- None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
A: Alignment, B: Robustness, C: Monitoring

17) "Deep RL methods outperform humans in simulated aerial combat."
What kind of hazard can this cause?

- CyberTerrorism
- Deception
- Weaponized AI
- Proxy gaming

No, the answer is incorrect.
Score: 0

Accepted Answers:
Weaponized AI

18) Which of these are not a part of Red Teaming a model?

- Checking the toxicity of the model by querying harmful content
- Checking the model generation speed to your queries
- Checking the bias in the model against certain genders and race
- Querying the model for your questions and doubts

No, the answer is incorrect.
Score: 0

Accepted Answers:
Checking the model generation speed to your queries
Querying the model for your questions and doubts

19) How do you measure the similarity between two protein structures?

- Local Distance Test
- Global Similarity Test
- Global Distance Test
- Local Similarity Test

No, the answer is incorrect.
Score: 0

Accepted Answers:
Global Distance Test

20) If the hardware access for AI building is limited to a few, it can cause severe inequality in which of the following

- Knowledge
- Economy
- Safety
- Power

No, the answer is incorrect.
Score: 0

Accepted Answers:
Economy
Power

21) "A team to identify vulnerabilities in AI systems within an organisation that builds AI systems."
The above defines which of the Organizational Risk's safety measures?

- Anomaly Detection
- Red Teaming
- Cyberattack
- Safety Testing
- Vulnerability Detection

No, the answer is incorrect.
Score: 0

Accepted Answers:
Red Teaming

22) What is the correct order of the steps involved in reducing the risk associated with AI systems.

- Respond -> Prioritize -> Identify -> Improve
- Prioritize -> Identify -> Improve -> Respond
- Respond -> Prioritize -> Identify -> Improve
- Identify -> Prioritize ->Respond -> Improve

No, the answer is incorrect.
Score: 0

Accepted Answers:
Identify -> Prioritize ->Respond -> Improve

23) Codex LM is used for which of the following?

- Convert code to its natural language description
- Convert math problems into step-by-step natural language explanation
- Convert natural language description of programming task to code
- Convert natural language explanations of math problems to their numerical equivalents
- Convert code from one programming language to the other

No, the answer is incorrect.
Score: 0

Accepted Answers:
Convert natural language description of programming task to code

24) Which of the following do you think are capable of discovering novel algorithms?

- Google's Minerva
- OpenAI's Codex
- Deepmind's AlphaFold3
- BigScience's BLOOM
- Deepmind's AlphaTensor
- StackOverFlow's OverflowAI

No, the answer is incorrect.
Score: 0

Accepted Answers:
Deepmind's AlphaTensor